



PG13E3G – 400 MHz Clock Generator

General Description

The PG13E3G is a general purpose, high-performance PLL-based clock generator.

It has a programmable output frequency, which ranges from 50 MHz to 400 MHz configured using a 5-bit input divider, an 8-bit feedback divider and a 2-bit output divider. It can also be used as a clock buffer through a bypass mode. A full power-down mode is also available.

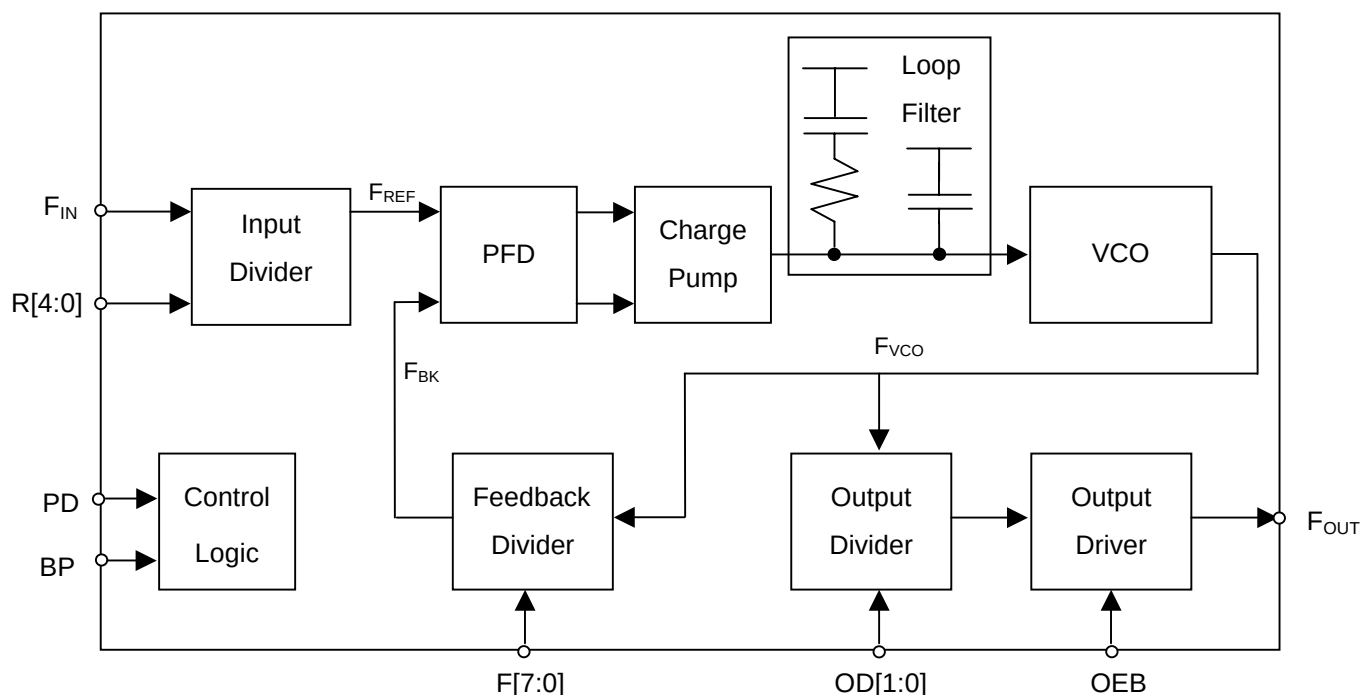
Technology

PG13E3G is a 1P4M layout structure based on TSMC 0.13um 1P4M~1P8M 1.2V/2.5V & 1.2V/3.3V logic process. In the layout of this cell, the via between metal-2 and metal-3 is of normal size. To fit the top-via specification of design rule which enlarge the via connecting to the top metal, we suggest our customer using PG13E3G in the chip which contains at least 4 layers of metal.

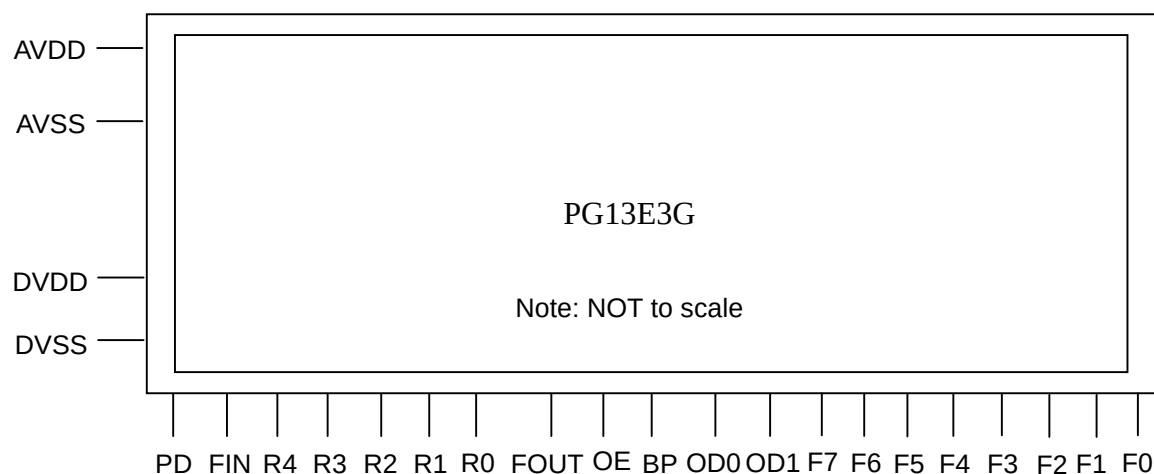
Features

- Fully integrated, including loop filter
- VCO operating range: 200 MHz – 400 MHz
- Output frequency range: 50 MHz – 400 MHz
- PFD comparison frequency range: 2 MHz – 8 MHz
- Contains 5-bit input, 8-bit feedback and 2-bit output dividers
- Input frequency range: 5 MHz – 250 MHz
- Input divider value: 1-31
- Feedback divider value: 26-200 (step by 2)
- Output divider value: /1, /2, /4
- Bypass mode
- Power-down mode
- Cell size: 448.98um x 445.14um
- Power supply: 1.2V single power supply

Block Diagram



Pin Placement / Definitions



PIN NAME	PIN TYPE	DESCRIPTION	INPUT CAPACITANCE (fF)
AVDD	P	Dedicated Analog power supply +1.2V	--
AVSS	G	Dedicated Analog ground	--
DVDD	P	Dedicated Digital power supply +1.2V	--
DVSS	G	Dedicated Digital ground	--
FIN	I	Input clock	30
FOUT	O	Adjustable clock output	--
OEB	I	FOUT enable pin	30
BP	I	Bypass the PLL	30
PD	I	Power down mode	30
OD1	I	Output divider control pin 1	30
OD0	I	Output divider control pin 0	30
R4	I	Input 5-bit divider control pin (MSB)	30
R3	I	Input 5-bit divider control pin	30
R2	I	Input 5-bit divider control pin	30
R1	I	Input 5-bit divider control pin	30
R0	I	Input 5-bit divider control pin (LSB)	30
F7	I	Feedback 8-bit divider control pin (MSB)	30
F6	I	Feedback 8-bit divider control pin	30
F5	I	Feedback 8-bit divider control pin	30
F4	I	Feedback 8-bit divider control pin	30

PIN NAME	PIN TYPE	DESCRIPTION	INPUT CAPACITANCE (fF)
F3	I	Feedback 8-bit divider control pin	30
F2	I	Feedback 8-bit divider control pin	30
F1	I	Feedback 8-bit divider control pin	30
F0	I	Feedback 8-bit divider control pin (LSB)	30

Specifications

Operating Conditions

PARAMETER	SYM	CONDITION	MIN	TYP	MAX	UNIT
Operating Voltage (Digital)	DVDD	--	1.08	1.20	1.32	V
Operating Voltage (Analog)	AVDD	--	1.08	1.20	1.32	V
Input Low Voltage	V _{IL}	--	0	--	0.2*DVDD	V
Input High Voltage	V _{IH}	--	0.8*DVDD	--	DVDD	V
Junction Temperature	T _J	--	-40	--	125	°C

Electrical Characteristics

Junction Temperature = -40 – 125°C, Operating Voltage = typical ± 10% (Unless otherwise specified)

PARAMETER	SYM	CONDITION		MIN	TYP	MAX	UNIT
Comparison Frequency	F _{REF}	F _{REF} = F _{IN} / NR		2	--	8	MHz
Input Clock Frequency ^[1,2]	F _{IN}	F _{IN} = NR * F _{REF}		5	--	250	MHz
VCO Operating Range	F _{VCO}	F _{VCO} = F _{IN} * NF / NR		200	--	400	MHz
Output Clock Frequency	F _{OUT}	F _{OUT} = F _{IN} * NF / (NR*NO)		50	--	400	MHz
Period Jitter (pk-pk) ^[3]	--	Clean Power	NO=1	--	--	100	ps
			NO=2	--	--	150	
			NO=4	--	--	200	
Cycle-to-Cycle Jitter (max) ^[3]	--	Clean Power	NO=1	--	--	50	ps
			NO=2	--	--	100	
			NO=4	--	--	150	

Long-Term Jitter (pk-pk) ^[3]	--	Clean Power Long-Term Delay Cycles=1024	NO=1	--	950 ^[4]	--	ps
			NO=4	--	850	--	ps
Duty Cycle	--			40	50	60	%
Pull-In Time + Lock-In Time	T _{RDY}			--	--	0.5	ms
Power Consumption (Normal Mode)	--	FVCO = 400MHz FOUT = 400MHz AVDD=DVDD=1.2V		--	2.5	--	mW
Power Consumption (Power-Down Mode)	--	PD=1 ,OEB=1 AVDD=DVDD=1.2V		--	<10	--	μW
F _{OUT} Rise / Fall Time ^[5]	T _{SR}	C _L : loading (fF) Simulation value	11.2+ 0.9562*CL @-40°, 1.32V, FF	15+ 1.2981*CL @25°, 1.2V, TT	21.3+ 1.8903*CL @125°, 1.08V, SS		ps

[1] We suggest our customer to put buffers in front of FIN pin to shape the input clock. Rise and fall time of the input clock being less than 1 ns is preferred.

[2] Please refer the **Basic Structure & Operation** section and our **Silicon Report** to see detail frequency relationship and divider setting.

[3] Please refer **Appendix A** to see the different jitter definitions. And given a jitter definition for measurement, the value is also different for different NF and NR divider's setting. Fin input jitter, divider setting, and power noise are the factors that affect the performance of PLL.

[4] Long-Term jitter's performance is strongly dependent on divider setting (loop bandwidth) and over how many cycles it should be measured. Please refer our **Silicon Report** to see the detail performance.

[5]: We define output slew rate as the output clock rising time delay from 0.1*DVDD to 0.9*DVDD (CL : Output Load in fF). The output loading condition should be carefully considered according to the clock output driving capability.

Operation

This macro cell has 4 modes of operation:

Normal Mode

The output clock frequency is programmable through the divider setting. It needs a **Tready time** (Pull_in Time + Locking Time) for PLL to re-lock the FIN clock when PLL wakes up from Power Down Mode operation to Normal Mode. In general, it should be reserved a **Tready** time for re-locking when PLL is changed to Normal Mode from Power Down or Bypass Mode, or when any divider setting is changed.

Bypass Mode

Bypass Mode provides FOUT with the same frequency as FIN.

Power-down Mode

Power Down Mode pulls FOUT to High and makes PLL in its low power consumption state.

Output Disable mode

While PLL loop is under normal operation, OEB can be set HIGH to disable output signal and Fout is set to HIGH.

Output Configuration

PD	BP	OEB	DESCRIPTION	F _{OUT}
1	x	x	Power Down Mode	1
0	1	0	Bypass Mode	FIN
0	0	0	Normal Mode	--
0	0	1	Output Disable mode	1

X: don't care

Implementing the Dividers

In *Normal Mode*, in order for the PLL to function properly, it is necessary to set suitable integer values for the dividers (NR for the input divider, NF for the feedback divider and NO for the output divider).

- Input Divider Value (**NR**)^[6]

$$NR = 16 \cdot R4 + 8 \cdot R3 + 4 \cdot R2 + 2 \cdot R1 + R0$$

- Feedback Divider Value (**NF**)^[7]

$$NF = 2 \cdot (128 \cdot F7 + 64 \cdot F6 + 32 \cdot F5 + 16 \cdot F4 + 8 \cdot F3 + 4 \cdot F2 + 2 \cdot F1 + F0)$$

- Output Divider Value (**NO**)

Output Divider Setting	NO	FOUT
OD0=0, OD1=0	Not allowed	--
OD0=1, OD1=0	1	FVCO
OD0=0, OD1=1 ^[8]	2	FVCO/2
OD0=1, OD1=1 ^[8]	4	FVCO/4

Programming the Output Clock Frequency

$$F_{REF} = F_{IN} / NR$$

$$F_{VCO} = F_{OUT} \cdot NO$$

$F_{OUT} = F_{IN} \cdot NF / (NR \cdot NO)$, where F_{REF} is the comparison frequency for the PFD.

For proper operation in normal mode, the following constraints *must* be satisfied:

$$2 \text{ MHz} \leq F_{REF} \leq 8 \text{ MHz}$$

$$200 \text{ MHz} \leq F_{VCO} \leq 400 \text{ MHz}$$

$$50 \text{ MHz} \leq F_{OUT} \leq 400 \text{ MHz}$$

Example I: To synthesize a 200 MHz output clock with an input frequency $F_{IN} = 15$ MHz.

- Set normal mode operation:
 $PD = 0, BP = 0, OEB = 0$
- Set **NR** to obtain F_{REF} within the PFD comparison frequency range:
Let $F_{REF} = 5$ MHz
Since $F_{REF} = F_{IN} / NR$,
 $NR = F_{IN} / F_{REF} = 15 \text{ MHz} / 5 \text{ MHz} = 3$
 $\rightarrow R[4:0] = NR = 3 = 00011_2$
- Set **NO** and ensure F_{VCO} within the VCO operating range:
Set $NO = 1$
 $F_{VCO} = F_{OUT} * NO$
 $F_{VCO} = 200 \text{ MHz} * 1 \rightarrow$ within the VCO operating range
 $\rightarrow OD[1:0] = 01_2$
- Set **NF** to obtain the F_{OUT} frequency:
 $NF = F_{OUT} * NR * NO / F_{IN}$
 $NF = 200 \text{ MHz} * 3 * 1 / 15 \text{ MHz} = 40$
 $\rightarrow F[7:0] = NF/2 = 20 = 00010100_2$

- Set **NF** to obtain the F_{OUT} frequency:

$$NF = F_{OUT} * NR * NO / F_{IN}$$

$$NF = 60 \text{ MHz} * 3 * 4 / 18 \text{ MHz} = 40$$

$$\rightarrow F[7:0] = NF/2 = 20 = 00010100_2$$

[6] Setting $[R4, R3, R2, R1, R0] = [0, 0, 0, 0, 0]$ is not allowed.

[7] Setting $[F7, F6, F5, F4, F3, F2, F1, F0] = [0, 0, 0, 0, 0, 0, 0, 0]$ is not allowed.

[8] When $[OD1, OD0] = [1, 0]$ or $[1, 1]$, F_{OUT} would have around 50% duty-cycle.

Example II: To synthesize a 60 MHz output clock with an input frequency $F_{IN} = 18$ MHz.

- Set normal mode operation:
 $PD = 0, BP = 0, OEB = 0$
- Set **NR** to obtain F_{REF} within the PFD comparison frequency range:
Let $F_{REF} = 6$ MHz
Since $F_{REF} = F_{IN} / NR$,
 $NR = F_{IN} / F_{REF} = 18 \text{ MHz} / 6 \text{ MHz} = 3$
 $\rightarrow R[4:0] = NR = 3 = 00011_2$
- Set **NO** and ensure F_{VCO} within the VCO operating range:
Set $NO = 4$
 $F_{VCO} = F_{OUT} * NO$
 $F_{VCO} = 60 \text{ MHz} * 4 \rightarrow$ within the VCO operating range
 $\rightarrow OD[1:0] = 11_2$

Power Pad Assignments

- PG13E3G requires two power pads for (**DVDD**, **AVDD**), and two ground pads for (**DVSS**, **AVSS**). Both (**DVDD**, **DVSS**) and (**AVDD**, **AVSS**) pairs should be considered as analog power. To Place & Route the PLL easily, all of them are defined as signals rather than global powers. We strongly recommend our customer giving them dedicated power pads rather than sharing them with the power (ground) of core logic. It could cause function failure at some logic patterns which have large number of MOS's switching if any power (ground) of PLL is shared by core logic.
- If pad counts or package pin counts are limited, two possible solutions are: (1). Dedicated pads and double-bonded to the same lead. (2) DVDD (DVSS) and AVDD (AVSS) share the same power pad.
- Do not share the power / ground package pins of this macro cell with the power / ground package pins of other core logic circuits.
- Please use the power cut cells (e.g., **PRDIODE8**) to separate the analog power & ground plane from digital power & ground plane at I/O area, you can use **PVDD1PC** pad for (**DVDD**, **AVDD**), **PVDD2PC** pad for pad ring power, **PVSS1P** pad for (**DVSS**, **AVSS**) and **PVSS2P** pad for pad ring ground. **PVDD2PC** is connected 1.2v power supply^[9].
- If the whole chip is not pad limited, please use more power pads with double (multiple)-bond configuration to reduce power (ground) bouncing due to inductance of bonding wires and fast clock switching.

Placement & Routing

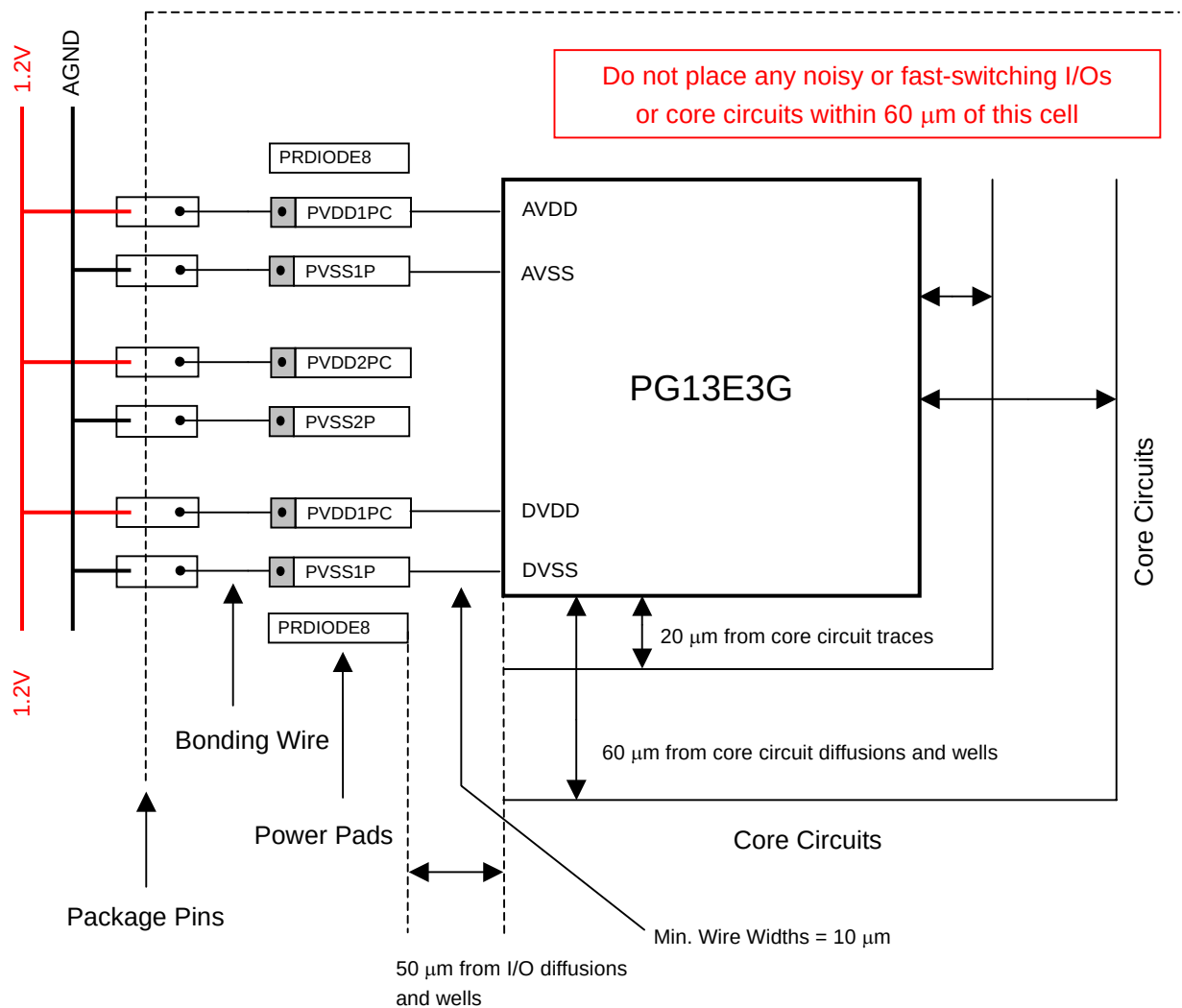
- We suggest that to place PG13E3G at corner, and keep the power and ground pins of PLL close to the analog power I/O pads. Keep the routing paths as short as possible and far away from any other high drive or simultaneously switching digital I/O pads.
- Don't place noisy or fast output drivers and potentially noisy internal circuits near the PG13E3G macro cell. In general, higher frequency signals generate more noise than lower frequency signals. Also, high-power level cells create more noise than low-power level cells.
- Core-logic power rings cannot cross over PG13E3G. Create dead space between the PG13E3G internal circuits and all other internal circuits. Physical space is a good way to reduce noise transmission through the chip substrate. It is recommended that a minimum 20um space between PG13E3G and digital core power rings is kept, and 60um space between PG13E3G and any other diffusion or well of digital core also.
- Avoid core-logic power lines and signal lines crossing over any power lines of PG13E3G.
- All power / ground pins of this macro cell are NOT global for LVS checking.

[9] Please refer to the "TSMC Analog I/O Application Notes" for update I/O naming and more detail information.

Placement & Routing

The figure below is a placement and routing example of PG13E3G macro. In this example, **PRDIODE8** power cut cells are inserted between power pads of PLL and other digital pads to isolate noise coupling. Double bonding is used at the power pad (**AVDD**, **AVSS**) to reduce the wire inductance. It is also suggested to minimize the bonding wire length of the power/ground pads if possible.

Example Place & Route



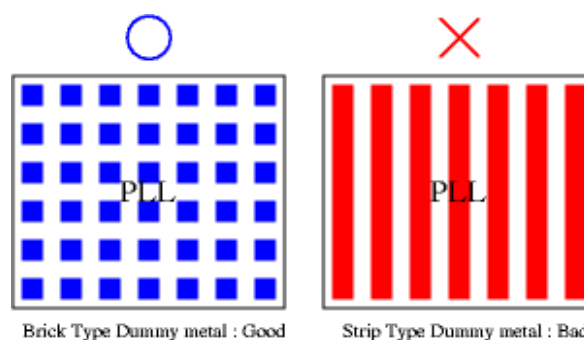
Placement & Routing

Treatment of metal density and signal routing above this macro cell

Basically, the PLL is 1P4M layout and the customers would put it into 1P4M ~1P8M full chip, thus there are metal density issue and routing possibility above the PLL if more than four metal layers are used. Since the vertical distance between M5/M6 and 1P4M PLL is short, the M5/M6 layers should be kept clean from signal routing path that either disturbs the PLL or adds parasitic capacitance on its internal nodes. The M7/M8 layers are allowed to use for routing. Following table describes the policy:

Customer's Layout layer used	Treatment for the metal layers above the PLL's layout			
	M5	M6	M7	M8
1P5M	Add dummy metal	N/A	N/A	NA
1P6M	Add dummy metal	Add dummy metal	N/A	NA
1P7M	Add dummy metal	Add dummy metal	Routing Allowed	NA
1P8M	Add dummy metal	Add dummy metal	Routing Allowed	Routing Allowed

There is also a preference for dummy metal pattern, as shown in the following graph. The little bricks act as isolated floating capacitors and do not affect the PLL, while the long strips provide coupling path and increase the parasitic capacitance at some internal nodes where speed is of concern.

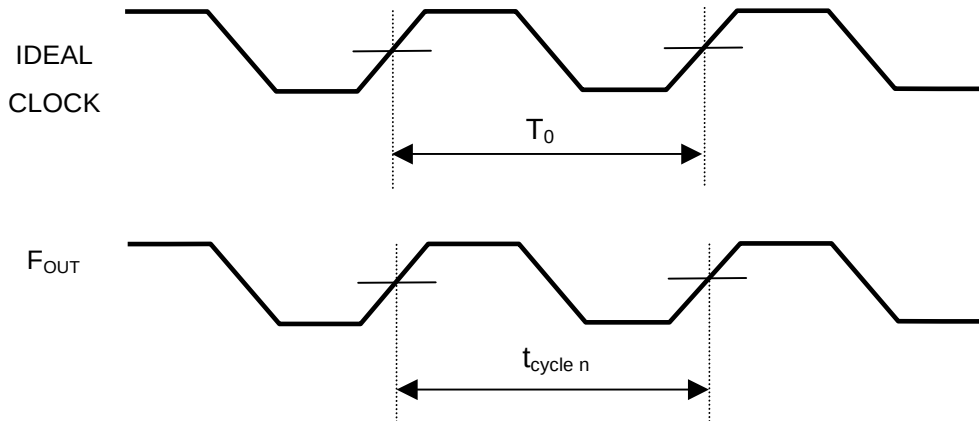


Appendix – Definitions of Terms

All definitions assume steady-state operation (T_{RDY} has been satisfied) and proper configuration of the PLL. The term Period Jitter, Cycle-to-Cycle Jitter, Static Phase Error, and Tracking Jitter are defined according to JEDEC standard No. 65-A (JESD65-A).

■ Period Jitter ($t_{jit(per)}$)

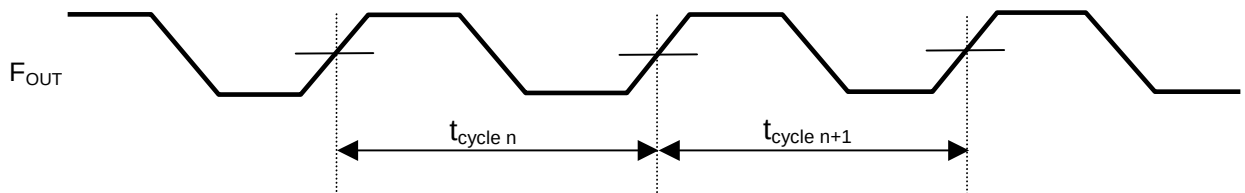
The deviation in cycle time of F_{OUT} with respect to the ideal period over a random sample of cycles.



$$t_{jit(per)} = |t_{cycle\ n} - T_0|, \text{ where } T_0 \text{ is the ideal output period and } t_{cycle\ n} \text{ is any cycle within the sample measured on controlled edges}$$

■ Cycle-to-Cycle Jitter ($t_{jit(cc)}$)

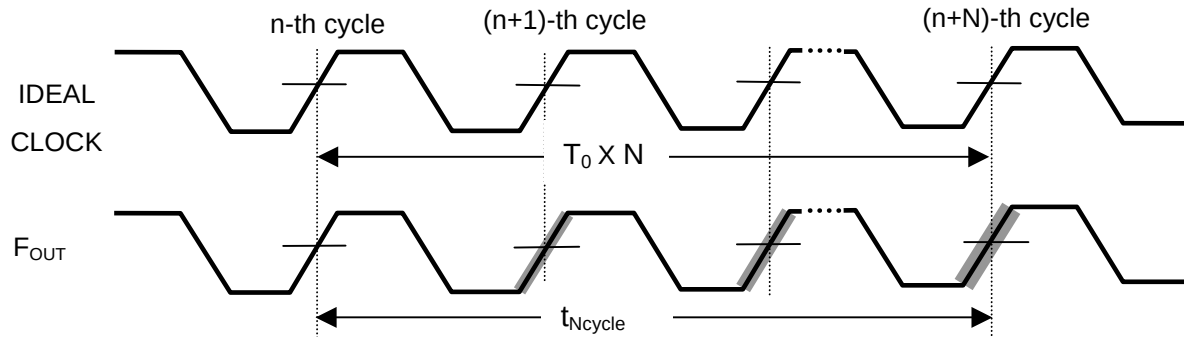
The deviation in cycle time of F_{OUT} between adjacent cycles, over a random sample of adjacent cycle pairs.



$$t_{jit(cc)} = |t_{cycle\ n} - t_{cycle\ n+1}|, \text{ where } t_{cycle\ n} \text{ and } t_{cycle\ n+1} \text{ are any two adjacent cycles measured on controlled edges}$$

■ Long-Term Jitter ($t_{jit(long-term)}$)

The edge deviation of F_{OUT} with respect to the ideal clock after a predetermined number of cycles.

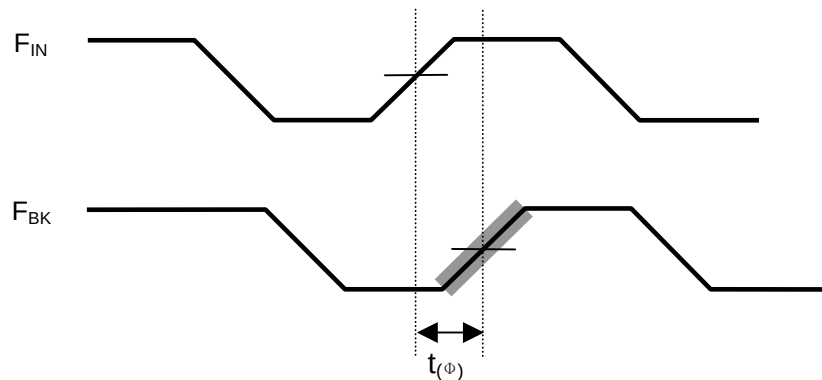


$$t_{jit(long-term)} = \left| t_{Ncycle} - T_0 \times N \right|, \text{ where } T_0 \text{ is the ideal clock period and a typical value of number } N \text{ is } 1024.$$

■ Static Phase Error ($t_{(\phi)}$)

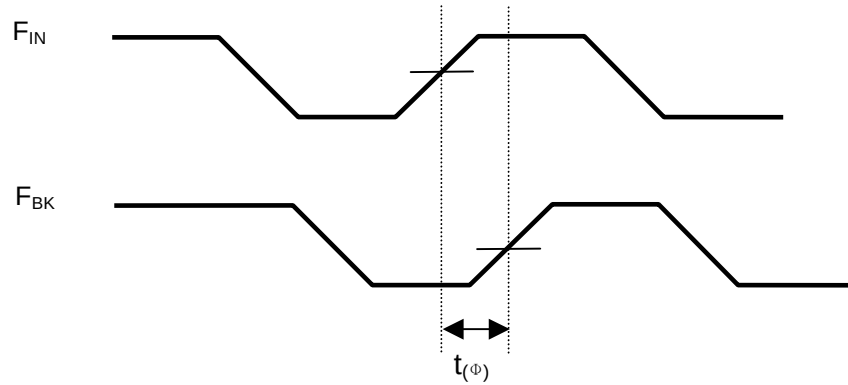
The time interval between similar points on the waveforms of the averaged input reference clock (F_{IN}) and the averaged feedback input signal (F_{BK}) when the PLL is locked and the input reference frequency is stable.

Static Phase Error > 0 means F_{BK} lags F_{IN} , and Static Phase Error < 0 means F_{BK} leads F_{IN}



■ Tracking Jitter ($t_{jit(\Phi)}$)

The deviation in $t_{(\Phi)}$ for a controlled edge with respect to a $t_{(\Phi)}$ mean in a random sample of cycles.



$t_{jit(\Phi)} = |t_{(\Phi)} - E(t_{(\Phi)})|$, where $t_{(\Phi)}$ is any random sample and $E(t_{(\Phi)})$ is the average of the sampled cycles measured on controlled edges.

■ Peak-to-Peak and RMS Values

The figure below shows the accumulated waveform and histogram of F_{OUT} rising edges. It was obtained from an oscilloscope with typical jitter measurement setup.

The peak-to-peak value is the difference between the maximum and minimum values of the histogram and the RMS value is the $1-\sigma$ (standard deviation) value of the distribution.

At least 20,000 samples are accumulated to obtain the histogram.

